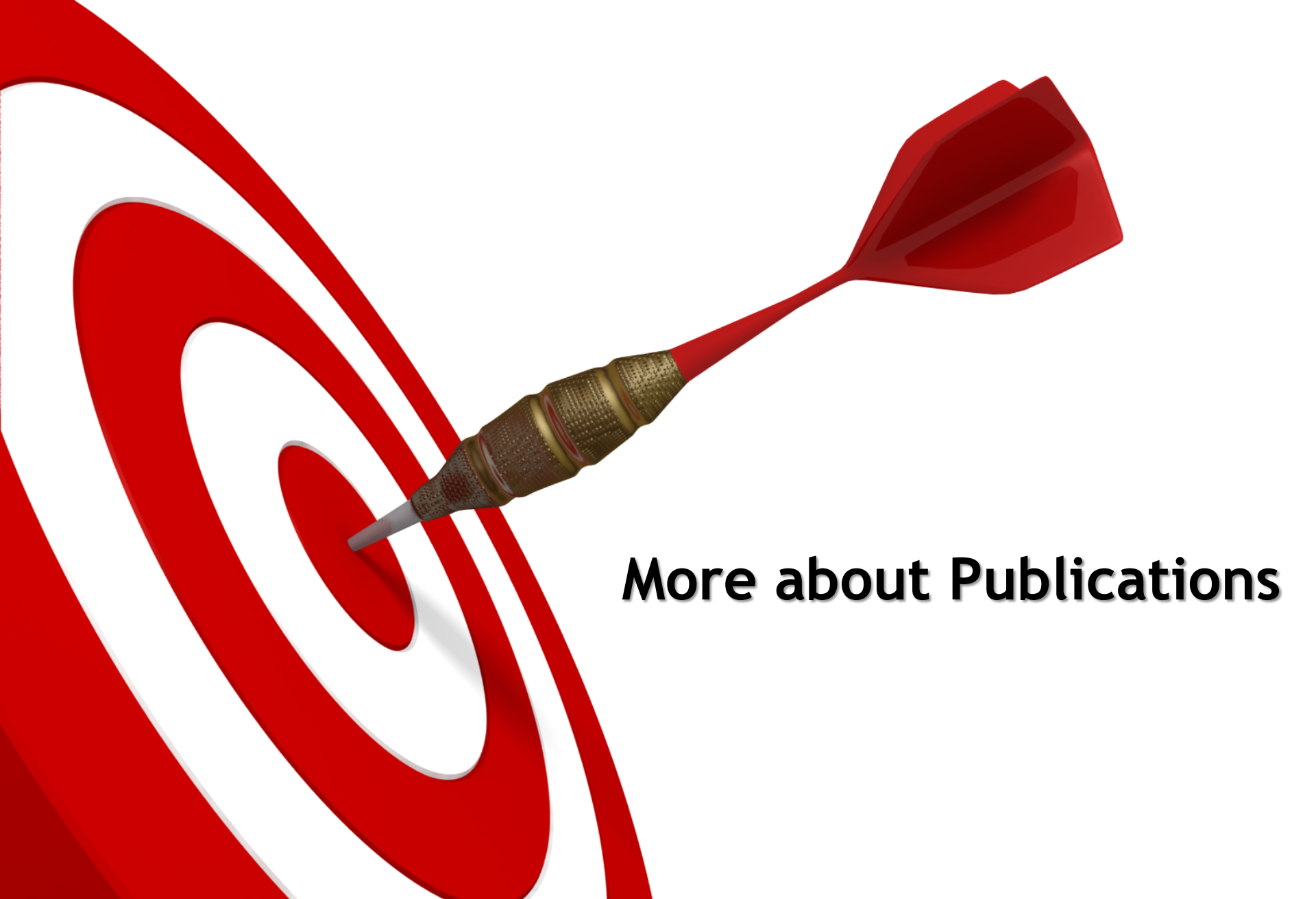




More about Publications

Types of Publications

- ☐ Theses/Dissertations
- ☐ Books
- ☐ Book Chapters (Handbooks)
- ☐ Research Reports
- ☐ Scientific Papers
 - Conferences
 - Scholarly Journals
 - Magazines
 - Surveys/Tutorials



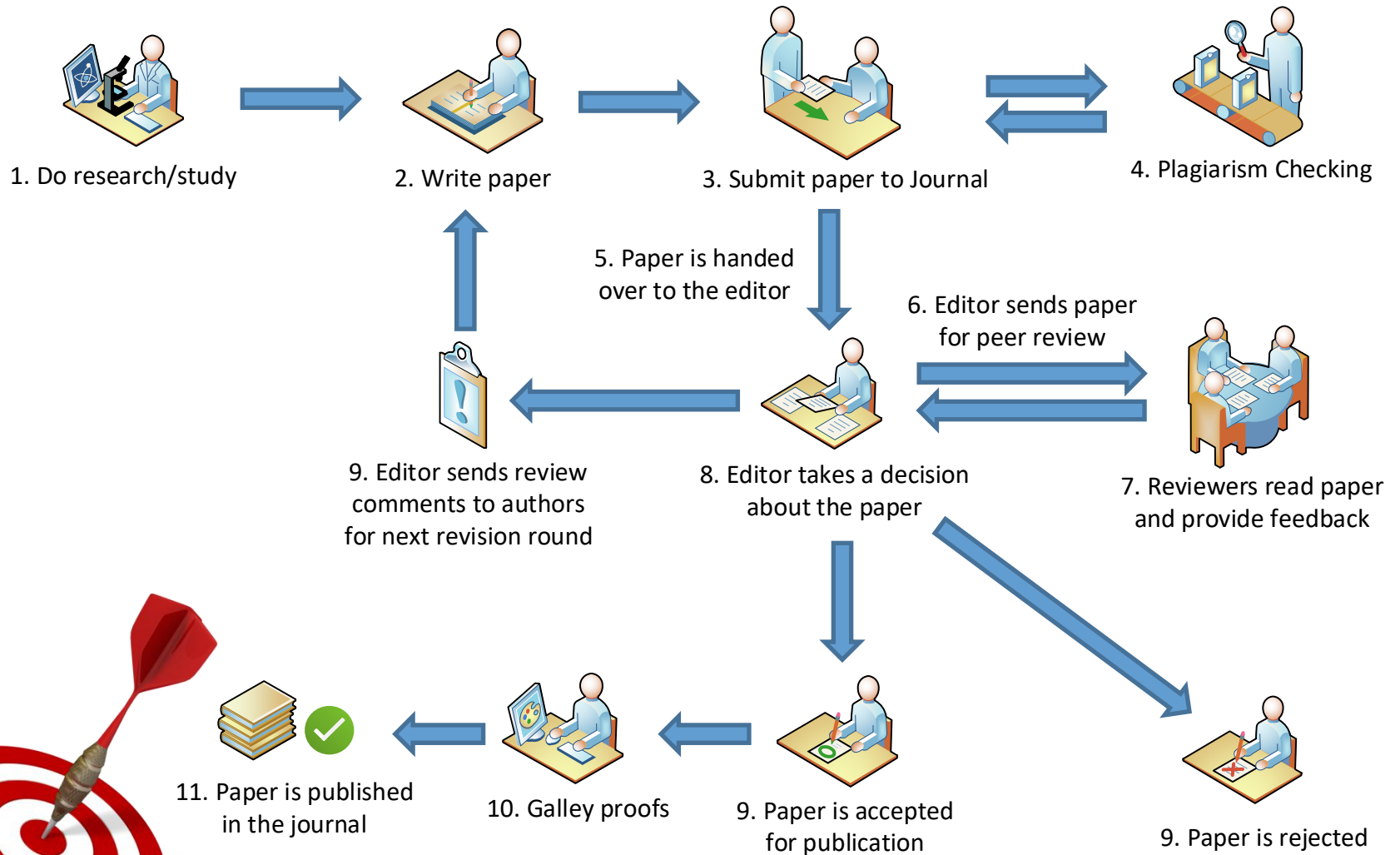
Review

- ❑ Academic publications are often examined before publication:
 - ❑ Dissertations are reviewed by an academic supervisor and opponent.
 - ❑ Books and book chapters are often reviewed by both an editor and the publisher.
 - ❑ Research reports are often inspected by colleagues within the same research field
 - ❑ Conference contributions are reviewed by those responsible for holding the conference (but also peer reviewed).
 - ❑ Scholarly journals are peer reviewed → the most comprehensive and esteemed version of review.



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Peer Review



Peer Review

- ❑ In order to avoid bias, a single-blinded or double-blinded review process is used:
 - Single-blind: the identities of the reviewers are not known to the authors, but the reviewers know the identities of the authors → **more traditional approach**
 - Double-blind: reviewers are not known to the authors and authors are not known to the reviewers



Listening to your reviewers

Every review like gold dust

Be (truly) grateful for criticism as well as praise

This is **really, really, really** hard

But it's
really, really, really, really, really, really, really, really,
really, really
important



Listening to your reviewers

- ☐ Read every criticism as a positive suggestion for something you could explain more clearly
- ☐ DO NOT respond “you stupid person, I meant X”. Fix the paper so that X is apparent even to the stupidest reader.
- ☐ Thank them warmly. They have given up their time for you.



Revision Example (Conference)

IEEE ICC 2018

Home Conference Travel grants My... Help

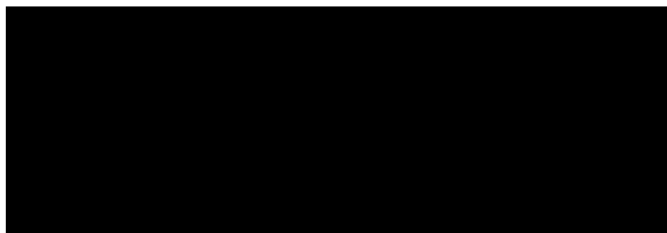
ICC'18 WCS #587 (1570407072): Efficient MIMO Detection for High-Order QAM Constellations in Time Dispersive Channels

#587 (1570407072): Efficient MIMO Detection for High-Order QAM Constellations in Time Dispersive Channels



Property	Change Add	Value																					
Conference and track		IEEE ICC 2018 Wireless Communications Symposium - IEEE ICC 2018 Wireless Communications Symposium																					
Authors		<table border="1"> <thead> <tr> <th>Name</th> <th>ID</th> <th>Edit</th> <th>Flag</th> <th>Affiliation (edit for paper)</th> <th>Email</th> <th>Country</th> </tr> </thead> <tbody> <tr> <td>Nuno Souto</td> <td>90778</td> <td></td> <td></td> <td>ISCTE/Instituto de Telecomunicações & Instituto de Telecomunicações, Portugal</td> <td>nuno.souto@lx.it.pt</td> <td>Portugal</td> </tr> <tr> <td>Rui Dinis</td> <td>975953</td> <td></td> <td></td> <td>Faculdade de Ciências e Tecnologia, University Nova de Lisboa & FCT-UNL, Portugal</td> <td>rdinis@fct.unl.pt</td> <td>Portugal</td> </tr> </tbody> </table>	Name	ID	Edit	Flag	Affiliation (edit for paper)	Email	Country	Nuno Souto	90778			ISCTE/Instituto de Telecomunicações & Instituto de Telecomunicações, Portugal	nuno.souto@lx.it.pt	Portugal	Rui Dinis	975953			Faculdade de Ciências e Tecnologia, University Nova de Lisboa & FCT-UNL, Portugal	rdinis@fct.unl.pt	Portugal
Name	ID	Edit	Flag	Affiliation (edit for paper)	Email	Country																	
Nuno Souto	90778			ISCTE/Instituto de Telecomunicações & Instituto de Telecomunicações, Portugal	nuno.souto@lx.it.pt	Portugal																	
Rui Dinis	975953			Faculdade de Ciências e Tecnologia, University Nova de Lisboa & FCT-UNL, Portugal	rdinis@fct.unl.pt	Portugal																	
Title	Only the chairs (icc18wcc-chairs@edas.info) can edit	Efficient MIMO Detection for High-Order QAM Constellations in Time Dispersive Channels																					
Abstract	Only the chairs (icc18wcc-chairs@edas.info) can edit	In this paper, we apply a generalized form of the alternating direction method of the multipliers (ADMM) and derive a multiple-input multiple-output (MIMO) detection algorithm for single carrier transmissions in time dispersive channels. The proposed algorithm supports different penalty parameters for each individual subcarrier and antenna and also includes a relaxation coefficient in the iterations. Besides evaluating the impact of these parameters, a method is presented for the automatic selection of the penalty. It is shown through simulations that very competitive performances can be obtained with the proposed approach for systems with high-order modulation combined with large antenna settings.																					
Topics	Only the chairs (icc18wcc-chairs@edas.info) can edit	MIMO, multi-user MIMO, and massive MIMO; Advanced equalization, channel estimation, and synchronization; Modulation, coding, and diversity techniques																					
Presenter(s)		presenter not specified																					
Session		WCS.23: Detection from Wed, May 23, 2018 16:15 until 18:00 (3rd paper) in New York B (17.5 min.)																					
DOI	Only the chairs (icc18wcc-chairs@edas.info) can edit																						
Status		Accepted																					
Visa letter		none																					
Review manuscript		However, authors cannot upload: paper status																					
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Personal notes																							

Revision Example (Journal)



Dear Prof. [REDACTED]

I write you in regards to manuscript [REDACTED], titled [REDACTED] Communications, which you submitted to IEEE Transactions on Wireless Communications.

In view of the criticisms of the reviewers found at the bottom of this letter and based on my own reading, your manuscript has been denied publication in the IEEE Transactions on Wireless Communications.

The topic and proposed detector are interesting. However, the paper lacks the novelties/contributions. It is a minor extension of its conference version [19]. Compared to [19], the additional technical material is not enough for this journal. In addition, the complexity is the main advantage of the proposed hybrid detection scheme, but there is no comprehensive complexity analysis. There are only some numerical results about the number of additions and multiplications with some specific setups.

Thank you for considering IEEE Transactions on Wireless Communications for the publication of your research. I hope the outcome of this specific submission will not discourage you from the submission of future manuscripts.

URL: <https://mc.manuscriptcentral.com/twc>

Sincerely,

Dr. [REDACTED]
Editor, IEEE Transactions on Wireless Communications

Reviewer(s)' Comments to Author:
Reviewer: 1

Comments to the Corresponding Author

The authors proposed a novel maximum likelihood (ML) detection algorithm for the MIMO system based on the hybrid operation of the sphere decoding (SD) and differential metrics (DM) on the tree search process, in which both the branch-and-bound principle and indicative functions are applied to remove the unnecessary search for reducing complexity. The proposed algorithm was shown to achieve ML performance with lower average complexity over the signal-to-noise ratio (SNR) regions. Although the proposed detector seems interesting, this reviewer raises major concerns about the technical contribution of the manuscript, because of the pretty old topic of MIMO detection with many papers in this area. This reviewer has the following major and minor comments:

- 1> In Eq. (7), the third line, the total number of the $\$j\$$ -th term should be $\$n-j+1\$$, rather than $\$n_j\$$
- 2> In Page 14, the formula is not well-organized, half of the formula is blocked.
- 3> MIMO detection is a rather old topic with many detection philosophies. For this reason, this reviewer thinks that the literature review is insufficient in the Introduction part and lacks an introduction to relevant detectors as well as recent progress on this topic.
- 4> This reviewer thinks that the simulation results are insufficient, the authors should add the complexity of other detectors, e.g., the full-search ML, successive interference cancellation (SIC) based detector, lattice reduction-based detector, etc., and compare with them.
- 5> As shown in Fig. 8, why does the proposed scheme have an increased complexity at the $\$SNR = -10\$$ dB.
- 6> Please elaborate on the effects of hyperparameters on the performance and complexity of the derived detector, e.g., the order of DM.
- 7> Please clarify the value of hyperparameters the authors adopted in the simulation.

1/2

This paper proposes a new ML detection algorithm for the MIMO system based on the hybrid operation of the modified sphere decoding (SD) algorithm and an algorithm based on differential metrics (DMs) on the tree search process, in which both the branch-and-bound principle and indicative functions are applied to remove unnecessary search.

Overall, the paper is well organized and written. However, there are several critical issues to be addressed.

1. This paper is extended from the conference work [19] published in 2018. It should be pointed out in the manuscript and the authors should clarify the differences between [19] and this paper. From my point of view, only Section III-D and III-E are new things, which seems to be a little stuffless to support another transaction paper.
2. The proposed hybrid algorithm can achieve much lower complexity below SNR = 0dB. It is better to add the related background about the range of SNR in actual application scenarios to demonstrate the practical value of the algorithm.
3. According to Figure 5, the algorithm combines SD and DM, so why is the complexity so much lower than both for low SNR ranges? It will be better to explain the reasons behind this situation in detail.
4. In Section IV, the simulation results of this paper, the numbers of antennas used are $n_T = n_R = 4$ and 5 respectively. Why choose the number of antennas with a relatively similar number? It would be better to choose the number of antennas with a larger gap, such as $n_T = n_R = 8$.
5. About "soft output", it needs to be verified in a coded MIMO system, as did in reference [17].
6. In the Abstract, Line 32, it will be easier for the reader to understand by changing "over signal-to-noise ratio (SNR)" into "over both the high and low SNR ranges".
7. In Section III, Page 6, Line 31, it writes that "have a left lower triangular matrix R", but in Section III-A, Page 7, Line 32, it writes "R is a right upper triangular matrix".
8. In Section III-B, as for DM-based ML Detection algorithm, formula derivation and many properties have been explained in detail in the reference paper [17], so it is not suitable to take up such a large space to introduce it.
9. In Section III-C, the paragraph starting at line 50 shows how the DM-based ML detection algorithm performs for the QAM constellation. It has little to do with the ML detection in hybrid mode and should be put into Section II-B.
10. How are the average numbers of additions and multiplications counted? Please specify it. Whether all computational complexity is included?
11. In Section III-E, Page 15, line 47, it says, "The soft output of $[S_m]_m$ in (42) can be approximated as $-[s_0]_m \delta(m)$ ". It would be helpful for the reader to give an explanation, or note that it is cited from reference paper [17].
12. In Fig. 6, the images on the left and right sides have different sizes.
13. In Fig. 9 and Fig. 10, the ordinate of the right images is incorrect. It should be "Average number of multiplications".

Exemplo 1

Exemplo 2

Revision Example (Journal) - Response

Physical Communication

Delay-Doppler Domain Decision Feedback Turbo Equalization for OTFS modulation
–Manuscript Draft–

Manuscript Number:	PHYCOM-D-21-00262R1
Article Type:	Full Length Article
Keywords:	OTFS; Delay-Doppler channels; Decision feedback equalization; Turbo equalization
Abstract:	Emerging evidence indicates that orthogonal time-frequency space (OTFS) modulation a potential candidate modulation scheme for high mobility wireless communications. However, OTFS may experience significant inter-symbol interference (ISI) and inter-Doppler interference (IDI) in the receiver. Here, we propose a soft decision feedback turbo equalization for OTFS transmission over delay-Doppler channels to jointly combat both interferences. A novel block decision feedback equalization (BDFE) algorithm is constructed using the band feature of the channels in the delay-Doppler domain. The feedforward and feedback filters are designed by the delay-Doppler channels coefficients. According to the designed filter, an equivalent system model is employed to allow turbo equalization. The posterior probability is established using the soft prior information and feedback filter, and then fed back to the channel decoder as external information. Both theoretical analysis and simulation results demonstrate the effectiveness of the proposed scheme in improving the bit error rate (BER) performance and combat various interference.

IEEE Wireless Communications Letters



**WIRELESS
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**Lightweight 1-D CNN-based Timing Synchronization for
OFDM Systems with CIR Uncertainty**

Journal:	IEEE Wireless Communications Letters
Manuscript ID	WCL2022-1306
Manuscript Type:	Original Article
Date Submitted by the Author:	17-Jul-2022
Complete List of Authors:	Qing, Chaojin; Xihua University School of Electrical Engineering and Electronic Information Tang, Shuhai; Xihua University School of Electrical Engineering and Electronic Information Cai, Xi; Xihua University School of Electrical Engineering and Electronic Information Wang, Jiafan; Xihua University School of Electrical Engineering and Electronic Information
Key Words:	Timing synchronization, Orthogonal frequency division multiplexing, Deep learning, Relaxed restriction for propagation delay

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Scientific Paper

- ☐ Presents the results and contribution of a research project following a scientific research methodology, carefully supported by literature review
- ☐ Presented to a scientific community
- ☐ Reviewed by the scientific community



Main Types of Papers

- ☐ Original Research Articles
- ☐ Theoretical articles
- ☐ Review articles
- ☐ Case Studies
- ☐ Methodologies or Methods



Original Research Article (Example)

- ❑ Report original analytical/experimental studies detailing the techniques and methods used.

IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE, VOL. 3, NO. 3, JUNE 2022

451

Robot Path Planning via Neural-Network-Driven Prediction

Jiankun Wang[✉], Jianbang Liu[✉], Weinan Chen, Wenzheng Chi[✉], and Max Q.-H. Meng[✉], *Fellow, IEEE*

Abstract—This article presents a novel heuristic method named *neural-network-driven prediction (NEED)* for robot path planning problems. Different from classical heuristic methods, NEED is not designed for some specific environments, which can be applied to various environments. NEED has an encoder-decoder structure, which consists of three modules: convolutional neural network backbone, spatial pooling module, and decoder module. By learning from a number of successful path planning cases, NEED can learn to analyze the environment structure and predict the promising search region for the new path planning problem. This predicted region serves as a heuristic to guide the search direction of path planning algorithms. A series of simulation experiments are conducted to demonstrate that NEED significantly improves the algorithm performance on solving new path planning problems. Meanwhile, NEED can also be applied to highly dynamic environments since it can output the prediction results at a speed of over 100 Hz on a laptop.

Index Terms—Artificial intelligence in robotics, intelligent robots, intelligent systems, neural networks, planning.

I. INTRODUCTION

PATH planning, as a fundamental problem in robotics [1], aims at finding a collision-free path that directs the robot to move from the start position to the goal position. It has been applied to many real-world applications, including but not limited to autonomous driving [2], humanoid robots [3], industrial manufacturing, and medical surgery. Many methods have been proposed to address the path planning problem, which can be generally classified into three categories. The grid-based methods discretize the whole state space into a number of grids and then implement the graph search algorithm like A* [4] to find a feasible path. However, they do not perform well in large-scale problems, and different parameter settings have a big impact on the planning result. The artificial potential field (APF) methods [5] use the attractive force from the goal position and the repulsive force from the obstacles to construct a potential field and then direct the robot to follow the steepest descent of the potential. But they suffer from the local minimum problem. The sampling-based methods such as probabilistic roadmap [6] and rapidly exploring random tree (RRT) [7] incrementally sample states from the state space to construct an implicit representation like a graph or tree, with which a feasible path can be found by connecting the start and goal node through edges. However, the generated path is nonoptimal, and the convergence speed to the optimal solution is slow.

To further improve the algorithm performance, a lot of heuristic methods are presented. They can be applied to either a certain environment or a certain algorithm. For example, in a sparse environment, adding more samples around the line connecting the start and goal position can dramatically reduce the planning time cost for the sampling-based algorithms. But this heuristic cannot be applied to other complex environments. Another example is using the relative distance between the robot and the goal to construct a new repulsive potential function in the APF algorithms [8], so that the problem of goals nonreachable with obstacles nearby can be avoided. Obviously, this heuristic cannot be applied to other path planning algorithms. In fact, the heuristic methods aim at providing an appropriate search direction for the path planning algorithm in the current environment. But this environment-related heuristic is very limited because the users do not know clearly that which heuristic should be used in the current environment, and the proposed heuristic cannot cover all

Manuscript received May 16, 2021; revised July 22, 2021 and August 31, 2021; accepted October 9, 2021. Date of publication October 15, 2021; date of current version May 25, 2022. This work was supported in part by the Shenzhen Key Laboratory of Robotics Perception and Intelligence under Grant ZDSY2020081017180001, in part by the Jiang Kong Innovation and Technology Commission (ITC) and Innovation and Technology Support Program Tier2 under Grant ITS210518P and in part by the Hong Kong JTC Midstream Research Program for Universities under Grant RMEDP0118. This paper was recommended for publication by Associate Editor Matthew Gerratt upon evaluation of the reviewers' comments. (Corresponding author: Max Q.-H. Meng.) Jiankun Wang and Weinan Chen are with the Department of Electronic and Electrical Engineering, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: wangjk@sustech.edu.cn; chenwn@sustech.edu.cn).

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Max Q.-H. Meng is with the Department of Electronic and Electrical Engineering, Southern University of Science and Technology, Shenzhen 518055, China, and also with Shenzhen Research Institute, The Chinese University of Hong Kong, Shenzhen 518172, China, on leave from the Department of Electronic Engineering, The Chinese University of Hong Kong, 999077, Hong Kong (e-mail: mengqh@sustech.edu.cn).

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Letters (Example)

- ❑ Letters communicate brief reports of data from original research that may stimulate further research in the field.
- ❑ Useful for scientists with results that are time sensitive.



Fast approach for efficient vehicle counting

M.A. Abdelwahab^a

Systems for counting vehicles should be fast enough to be implemented in real-time situations. Most of the related work uses two stages for vehicle counting, vehicle detection and tracking, which increase the computational complexity. In this Letter, a fast and efficient approach for vehicle counting is proposed, where there is no need for the vehicle tracking step. A background model is created only for a narrow region, a line, in the video frames. The moving vehicles are detected as foreground objects while passing the narrow region. Morphological processes are applied to the extracted objects to enhance them and decrease the effects of vehicle occlusions. Finally, an efficient counting vehicle method is introduced employing only the extracted detection information. The experimental results performed on diverse videos show that the proposed method is fast and accurate. The average execution time per frame is 7.78 ms.

Introduction: Counting vehicles is a very essential issue due to its importance in many traffic applications [1, 2]. Both vehicles detection and tracking are often necessary for vehicle counting. Generally, vehicle detection techniques can be classified into appearance based and motion based [3]. In the appearance-based techniques, features such as colour, edge and contour are used to detect vehicles. In the motion-based techniques, vehicles are detected employing their motion such as methods based on background subtraction, frame differences and optical flow. Due to its efficiency, background subtraction is a very popular technique in vehicle counting. So, many methods used it [4]. In these methods, a background model is built, then the background image is subtracted from every frame to obtain the foreground objects. However, in most of these methods, the background model is created for the whole frame or for a large area in it. Also, a vehicle tracking step is usually used after vehicle detection. Kalman and particle filters [1, 4] or feature tracking [5] can be used for tracking. Therefore, these methods almost have a high computational complexity. In this Letter, a new counting vehicle method based on background subtraction is proposed. In this method, the background model is only created for a narrow region in the frame (it is shown as a red line in Fig. 1). Then, vehicles are detected and counted when passing through this line. A way for non-repeated vehicle counting is introduced, where there is no need for a tracking step. So, the algorithm complexity is decreased and the counting process becomes very fast.

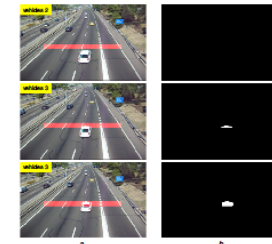


Fig. 1 Example for counting vehicles using only the extracted foreground objects information. ROI is highlighted by red colour
a From top to bottom represents three consecutive frames from M-30 video
b Corresponding foreground objects

Foreground object extraction: The background model is created for a small area, the region of interest (ROI), as shown in Fig. 1. Therefore, the foreground objects extraction process is fast. The Gaussian

mixture modelling (GMM) is used for modelling the background, where each background pixel is modelled by a mixture of K Gaussian distributions. To make the system robust to the background changes, the background model is reset every M frames, i.e. it is instantiated to build itself. To avoid an abrupt change in the foreground detection during the reset moment, two identical models are created. They work alternately with a shift of $M/2$ frames, i.e. when one model is reset, the other model is working for $M/2$ frames and so on.

Foreground object refinement: The extracted regions have noisy objects, which have small areas. Also, there are a lot of holes in the extracted areas. These problems could cause a false detection. To overcome these problems, refinement processes are applied. First, the small objects are removed. Then, morphological processes are applied to the remaining objects. These processes consist of dilation followed by erosion. The purpose of dilation is to close the holes in the extracted objects, while erosion is used to decrease the object boundaries resulted from dilation. Generally, the amount of dilation areas equal to those of erosion areas, but in our method the erosion could be used by an amount more than that of dilation. The reason is to decrease the common area between adjacent vehicles, especially for crowded scenes.

Vehicle counting: The most important step is how to convert the extracted objects into vehicle count information without any tracking technique. When a vehicle passes the ROI at consequent frames, a part of the vehicle is detected as a foreground object in every frame. The problem is how to count these objects as one vehicle without repetition. To perform that, every detected foreground region in each frame takes a number N_f , where f and v represent the frame number and the vehicle label within this frame (1: n), respectively. Then, each of these regions is compared with those n detected regions in the previous frame N_{f-1} , where $u = 1:n$. This comparison is based on how much is the common area between two objects in two consecutive frames. If the common area is greater than a predefined threshold, it is considered that these objects belong to one vehicle. In this case, the object in the frame f takes the same vehicle label of the matched object in the frame $f-1$, i.e. $N_f = N_{f-1}$, otherwise it takes a new label and the vehicle number is incremented by one.

Fig. 1 shows an example for counting vehicles using only the foreground object detection information at consequent frames. The figure represents the moving of a car at three consequent frames from top to bottom. At the second frame (second row), the counter is increased by one due to detection of a new foreground object, which does not match any object at the first frame (first row). The counter is not changed in the third frame because the detected foreground object is matched with the one at the previous frame.

Experimental results and analysis: To evaluate the proposed method, three experiments were employed using five videos representing different circumstances. In all experiments, the GMM was used for creating the background model for only the ROI. Two identical models were created, they worked alternately with a shift of $M/2$ frames, where $M = 50$ frames. Each model was reset every M frames. The number of Gaussians (K) for each model was selected to be five to make the background model more robust to changes. The values of M and K were determined empirically. For enhancing the extracted foreground, a closing morphological operation (dilation followed by erosion) was applied using a square structuring element with a size of 3×3 pixels. An extra erosion step, 15×15 pixels, was used to decrease the effect of vehicle occlusions.

In experiment 1, two videos from the GRAM dataset [6] were used. They are M-30 and M-30 HD. Samples of counting results for both videos are shown in Fig. 2. The proposed method was compared to state-of-the-art methods which used those videos as shown in Table 1. The precision was used as an accuracy measurement. It is defined as in equation (1).

$$\text{Precision}(\%) = 100 - \text{Error}(\%)$$

where

$$\text{Error}(\%) = \frac{|\text{Estimated} - \text{True No.}|}{\text{True No.}} \times 100 \quad (1)$$

The results of the compared methods were taken from [1]. The real vehicle numbers used by these methods were 77 and 42 vehicles in

Theoretical Article (Example)

- ❑ Introduces and discusses abstract ideas and principles, either new or established, that are related to a specific field of study or body of knowledge.
 - aimed at explaining, predicting and understanding phenomena.



Appl. Comput. Harmon. Anal. 35 (2013) 74–93



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Sharp RIP bound for sparse signal and low-rank matrix recovery

T. Tony Cai^{a,*}, Anru Zhang

University of Pennsylvania, United States

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 Nuclear norm minimization
 Restricted isometry
 Sparse signal recovery

ABSTRACT

This paper establishes a sharp condition on the restricted isometry property (RIP) for both the sparse signal recovery and low-rank matrix recovery. It is shown that if the measurement matrix A satisfies the RIP condition $\delta_k^4 < 1/3$, then all k -sparse signals β can be recovered exactly via the constrained ℓ_1 minimization based on $y = A\beta$. Similarly, if the linear map M satisfies the RIP condition $\delta_k^4 < 1/3$, then all matrices X of rank at most r can be recovered exactly via the constrained nuclear norm minimization based on $b = M(X)$. Furthermore, in both cases it is not possible to do so in general when the condition does not hold. In addition, noisy cases are considered and oracle inequalities are given under the sharp RIP condition.

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1. Introduction

Compressed sensing has been a very active field of recent research with a wide range of applications, including signal processing, medical imaging, seismology, and statistics. The goal is to develop efficient data acquisition techniques that allow accurate reconstruction of highly undersampled sparse signals. It is now well understood that the constrained ℓ_1 minimization method provides an effective way for recovering sparse signals. See, e.g., Candès and Tao [6,7], Donoho [11] and Donoho, Elad, and Temlyakov [12]. A closely related problem is the affine rank minimization problem, where the goal is to recover a large low-rank matrix based on an observation of an affine transformation of the matrix. Applications include linear system identification and control, Euclidean embedding, and image compression. See, e.g., Candès and Plan [9], and Recht, Fazel and Parrilo [18].

More specifically, in compressed sensing, one observes (A, y) with

$$y = A\beta + z \quad (1)$$

where $y \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times p}$ with $n < p$, $\beta \in \mathbb{R}^p$ is a sparse signal of interest, and $z \in \mathbb{R}^n$ is a vector of measurement errors. One wishes to recover the unknown sparse signal $\beta \in \mathbb{R}^p$ based on A and y using an efficient algorithm. The affine rank minimization problem aims to reconstruct a low-rank matrix X based on a known linear map M and an observed vector $b \in \mathbb{R}^n$ where

$$b = M(X) + z. \quad (2)$$

^{*} Corresponding author.
 E-mail address: cai@wharton.upenn.edu (T.T. Cai).
¹ The research of Tony Cai was supported in part by NSF FRG Grant DMS-0854873.

1063-5203/\$ – see front matter © 2012 Published by Elsevier Inc.
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Case Study (Example)

❑ Report specific instances of interesting phenomena

IEEE INTERNET OF THINGS JOURNAL, VOL. 9, NO. 16, 15 AUGUST 2022

15161

Dealing With Challenges Developing a Gateway to the “End of the World”—The Brazilian Antarctic Station Case Study

Daniela Pawelski Amaro Marins[✉], Jordano R. Celestrini[✉], Vinícius Ferreira de Almeida, Celso Alberto Saibel Santos, Rodrigo V. Andréo, Roberto Colistete Júnior, Cristina Engel de Alvarez, and Marcelo Eduardo Vieira Segatto

Abstract—The circadian rhythm is the period of about 24 h, which is the basis of the human biological cycle. It coordinates human daytime activities and body rest, and influences cognitive learning and processing information in the brain during sleep. Light rhythmically influences these hormonal physiological procedures. External factors such as the exposure to artificial light can also interfere with the circadian rhythm, causing sleep and wake disturbances. Remote monitoring systems can be used to better understand how the artificial light influences sleep and bodily rhythms, integrating light and physiological measurements to assess their interaction points. We developed an equipment to monitor the environmental light spectrum and gather data from wrist devices, collecting data on sleep quality and daytime activities. This study shows the challenges faced during the development of this monitoring equipment and the lessons learned from its construction and installation at the Brazilian Antarctic research station.

Index Terms—Antarctica, circadian cycle, IoT, photonics, remote patient monitoring (RPM).

I. INTRODUCTION

HUMAN beings are increasingly dependent on light to survive. Contemporary life requires 24-h illuminated

environments: besides daytime activities, we also use light for nighttime activities and even to sleep. Concerns about the consequences of this uninterrupted stimulus are both already evident and a research priority of the International Commission on Illumination (CIE) [1]. Light directly influences the human metabolism due to exclusive receptor glands in the human eye controlling the production of light-regulated melatonin which guides human rest and activity [2], [3].

Research on this topic is quite diverse and concerns the use of artificial light during the day to replace the sun in addition to other equipment emitting artificial light, which belongs to human routine, such as artificial light itself, the illuminated screens of cell phones, TVs, and computers [4].

Since light plays an important role in how our body works, scientists have made efforts, in recent years, to best analyze and measure this influence. The literature reports tests with a variety of subjects, such as starlight analysis, digital monitoring, questionnaires, urban daily life, and architecture [5]–[8]. However, all studies agree on one key point: light is the main external agent responsible for suppressing melatonin and, consequently, essential for human rest and the maintenance of circadian cycles [9]–[11].

A possible strategy to assess the influence of light on sleep quality is to monitor people and environment. In healthcare, remote patient monitoring (RPM) has gained notoriety, proposing the use of sensors to monitor vital signs, such as electrical signals, heart rate, respiratory rate, blood pressure, temperature, glucose levels, physical activity, and sleep quality [12]. Some RPM platforms also can gather environmental data, such as temperature, lighting, and carbon dioxide (CO₂) concentrations, enabling to assess the correlation between environmental variables and human physiological data. This study encompasses these innovative technologies to monitor the environmental light and physical human activity to evaluate the relation between these variables.

Choosing an isolated location to remotely monitor participants may reduce the number of variables and minimize the impact of other elements. It may also challenge participants' behavior who, living away from family and friends, need physiological regularity to supply their social and communication needs. Several studies have already explored how RPM systems can monitor participants in remote regions, mainly in rural areas [13]–[15]. However, applying these systems

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Research Ethics Committee under Application No. 232348/19.5.0000.5542 (CEP/UFES), and performed in line with the National Health Council of Brazil.

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Methodologies Article (Example)

- ❑ Present a new experimental method, test or procedure.



Review Article (Literature Review Example)

- ❑ Evaluation of published literature through comparison, analysis, pointing issues, or offering tips for future studies.

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IEEE COMMUNICATIONS SURVEYS & TUTORIALS, VOL. 24, NO. 3, THIRD QUARTER 2022

Internet of Intelligence: A Survey on the Enabling Technologies, Applications, and Challenges

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Abstract—The Internet of Intelligence is conceived as an emerging networking paradigm, which will make intelligence as easy to obtain as information. This paper provides an overview of the Internet of Intelligence, focusing on motivations, architecture, enabling technologies, applications, and existing challenges. This can provide a good foundation for those who are interested to gain insights into the concept of the Internet of Intelligence and the key enablers of this emerging networking paradigm. Specifically, this paper starts by investigating the evolution of networking paradigms and artificial intelligence (AI), based on which we present the motivations of the Internet of Intelligence by demonstrating that networking needs intelligence and intelligence needs networking. We then present the layered architecture to characterize the Internet of Intelligence systems and discuss the enabling technologies of each layer. Moreover, we discuss the critical applications and their integration with the Internet of Intelligence paradigm. Finally, some technical challenges and open issues are summarized to fully exploit the benefits of the Internet of Intelligence.

Index Terms—Internet of Intelligence, artificial intelligence, networking paradigm, architecture, enabling technologies, applications, challenges.

I. INTRODUCTION

WITH decades of research and development, the Internet has become an essential information infrastructure in today's world to facilitate economic development and social progress. According to Cisco's white paper, by 2023, the total number of Internet users will reach 5.3 billion, and networked devices will exceed 29.3 billion [1]. Through enabling

information networking among people and machines, the Internet can instantly transmit information to people thousands of miles away, realizing global informatization. Despite these benefits, there are still some new challenges in the post-Internet era: i) Information explosion: information overload may occur due to the ever-increasing information in the Internet, which will lead to difficulty in decision-making [2]; ii) Fake information: more and more fake information and other lousy information spread rapidly and widely through the Internet, which destroys the harmonious environment of the Internet and harms the economy and society [3]; iii) Human-in-the-loop: the explosive trend of the Internet has brought unprecedented challenges of scale, complexity, dynamics and cost to the current "human-in-the-loop" network operation and management [4]; iv) Limitation: the Internet exhibits limitations in solving existing challenges, including developing reliable, cost-effective autonomous systems (for example, autonomous driving).

To cope with these challenges, networking issues need to be considered in a larger timescale. Networking is for obtaining not just information but also matter and energy. Reviewing the development history of human society, we can find that cooperation is the heart. As a social species, human beings rely on cooperation to achieve survival and prosperity. Consequently, in modern history, to promote human cooperation in the socio-economic system, humans have developed technologies to enable networking for matter (grid of transportation), for energy (grid of energy), and for information (the Internet) [5]. In every evolution of the networking paradigm, we can move "something" (such as matter, energy, or information) to reduce the disparity of "something", thereby facilitating human cooperation through sharing "something". In such a large time scale, we can observe the evolution pattern of networking paradigm: the new networking paradigm is based on the existing paradigm, but provides a higher level of abstraction.

The future networking paradigm is inseparable from intelligence to solve the problems existing in the current information networking paradigm. Fig. 1 shows a conceptual relationship among data, information, and intelligence. Intelligence is the further abstraction and concentration of information [6]. Although much progress has been made in developing artificial intelligence (AI) [7]–[9], it is far from human learning, which requires much fewer data sets and is more flexible in adapting to new environments. According to the Big History Project [10], a decisive feature of human learning is collective learning. Through collective learning, human beings can

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Review Article (Systematic Review Example)

- ❑ Synthesizes findings in published literature, qualitatively or quantitatively.



Understanding Online Privacy—A Systematic Review of Privacy Visualizations and Privacy by Design Guidelines

SUSANNE BARTH, DAN IONITA, and PIETER HARTEL, University of Twente, Netherlands

Privacy visualizations help users understand the privacy implications of using an online service. Privacy by Design guidelines provide generally accepted privacy standards for developers of online services. To obtain a comprehensive understanding of online privacy, we review established approaches, distill a unified list of 15 privacy attributes and rank them based on perceived importance by users and privacy experts. We then discuss similarities, explain notable differences, and examine trends in terms of the attributes covered. Finally, we show how our results provide a foundation for user-centric privacy visualizations, inspire best practices for developers, and give structure to privacy policies.

CCS Concepts: • Security and privacy → Usability in security and privacy; Privacy protections, Social aspects of security and privacy;

Additional Key Words and Phrases: Privacy attributes, privacy factors, privacy labels, privacy icons, privacy by design

ACM Reference format:

Susanne Barth, Dan Ionita, and Pieter Hartel. 2022. Understanding Online Privacy—A Systematic Review of Privacy Visualizations and Privacy by Design Guidelines. *ACM Comput. Surv.* 55, 3, Article 63 (February 2022), 37 pages.

<https://doi.org/10.1145/3502288>

1 INTRODUCTION

Online services currently handle unprecedented amounts of user-related data [129]. Machine learning algorithms extract value from large amounts of data by recognizing hidden patterns, links, behaviors, trends, identities, and practical knowledge, which has given birth to a “big data economy” [9, 152]. This has opened a “Pandora’s Box” of privacy concerns [113, 141, 151]. But the privacy policies that are meant to address these concerns are often lengthy, legally worded documents written to protect the provider [15, 59]. Even the interactive permission system found on modern smartphones fails to provide a sufficient understanding of the privacy risks involved with using an application [24, 39, 78].

To communicate privacy risks to users in a clear and concise manner, researchers, regulators, and industry have called for a more visual representation of how online services handle personal

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<https://doi.org/10.1145/3502288>

ACM Computing Surveys, Vol. 55, No. 3, Article 63. Publication date: February 2022.

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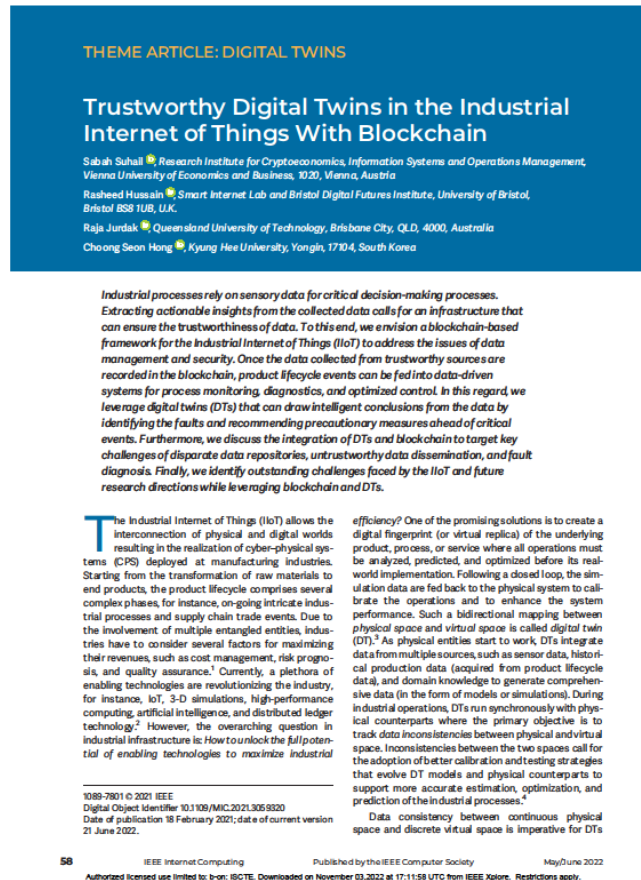
Review Article - Journals

- ❑ Some journal are dedicated solely to this type of articles:
 - ACM Computing Surveys
 - IEEE Communications Surveys and Tutorials
 - Proceedings of the IEEE
 - Computer Science Review (Elsevier)



Magazine (Example 1)

- Articles are tutorial in nature and written in a style comprehensible to professionals within the field that are working outside the specialty of the article.



Magazine (Example 2)

□ Magazine article versus research article on the same subject.



The authors provide an overview of the IRS technology, including its main applications in wireless communication, competitive advantages over existing technologies, hardware architecture as well as the corresponding new signal model. They address the key challenges in designing and implementing the new IRS-aided hybrid wireless network, as compared to the traditional network comprising active components only.

ABSTRACT

IRS is a new and revolutionizing technology that is able to significantly improve the performance of wireless communication networks, by smartly reconfiguring the wireless propagation environment with the use of massive low-cost passive reflecting elements integrated on a planar surface. Specifically, different elements of an IRS can independently reflect the incident signal by controlling its amplitude and/or phase and thereby collaboratively achieve fine-grained 3D passive beamforming for directional signal enhancement or nulling. In this article, we first provide an overview of the IRS technology, including its main applications in wireless communication, competitive advantages over existing technologies, hardware architecture as well as the corresponding new signal model. We then address the key challenges in designing and implementing the new IRS-aided hybrid (with both active and passive components) wireless network, as compared to the traditional network comprising active components only. Finally, numerical results are provided to show the great performance enhancement with the use of IRS in typical wireless networks.

INTRODUCTION

The targeted 1000-fold network capacity increase and ubiquitous wireless connectivity for at least 100 billion devices by the forthcoming fifth-generation (5G) wireless network have been largely achieved, thanks to the various key enabling technologies such as ultra-dense network (UDN), massive multiple-input multiple-output (MIMO), millimeter wave (mmWave) communication, and so on [1]. However, the required high complexity and hardware cost as well as increased energy consumption are still crucial issues that remain unsolved. For instance, densely deploying base stations (BSs) or access points (APs) in a UDN not only entails increased hardware expenditure and maintenance cost, but also aggravates the network interference issue. In addition, extending massive MIMO from sub-6 GHz to mmWave frequency bands generally requires more complex signal processing as well as more costly and energy-consuming hardware (e.g., radio frequency (RF) chains). Therefore, research on finding

innovative, spectral and energy efficient, and yet cost-effective solutions for future/beyond-5G wireless networks is still imperative [2].

Recently, intelligent reflecting surface (IRS) has been proposed as a promising new technology for reconfiguring the wireless propagation environment via software-controlled reflection [3–6]. Specifically, IRS is a planar surface comprising a large number of low-cost passive reflecting elements, each being able to induce an amplitude and/or phase change to the incident signal independently, thereby collaboratively achieving fine-grained three-dimensional (3D) reflect beamforming. In a sharp contrast to the existing wireless link adaptation techniques at the transmitter/receiver, IRS proactively modifies the wireless channel between them via highly controllable and intelligent signal reflection. This thus provides a new degree of freedom (DoF) to further enhance the wireless communication performance and paves the way to realize a smart and programmable wireless environment. Since IRS eliminates the use of transmit RF chains and operates only in short range, it can be densely deployed with scalable cost and low energy consumption, yet without the need of sophisticated interference management among passive IRSs. Furthermore, IRSs can be practically fabricated to conform to mount on arbitrarily shaped surfaces to cater to different application scenarios, while the underlying communication modeling and problem need further investigation.

Figure 1 illustrates several typical applications of IRS-aided wireless networks. The first application considers a user located in a dead zone where the direct link between it and its serving BS is severely blocked by an obstacle. In this case, deploying an IRS that has clear links with the BS and user helps bypass the obstacle via intelligent signal reflection, thereby creating a virtual line-of-sight (LoS) link between them. This is particularly useful for the coverage extension in mmWave communications that are highly vulnerable to indoor blockage. The second application shows the use of IRS for improving the physical layer security. When the link distance from the BS to the eavesdropper is smaller than that to the legitimate user (e.g., user 1), or the eavesdropper lies in the same direction as the legitimate user (e.g., user 2), the achievable secrecy commun-

A Low Complexity Algorithm for Achievable Rate Maximization in mmWave Systems aided by IRS

Mingli Shi, Xiaohui Li, Member, IEEE, Tao Fan, Jiawen Liu and Siting Lv

Abstract—This letter investigates the weighted sum-rate maximization in a multi-user millimeter wave (mmWave) wideband downlink system aided by double intelligent reflecting surfaces (IRS). We formulate a joint optimization problem of digital beamforming (BF) matrix, analog BF matrix, and IRS phase shifts matrix, which is non-convex. To solve this problem, we propose an alternating optimization (AO) algorithm based on the projection gradient method (PGM). PGM is used to deal with non-convex element-wise constant-modulus constraints for analog BF matrix and IRS phase shifts. Furthermore, the performance loss caused by using discrete phase shift is studied by the proposed algorithm and quantization method. Simulation results show that the proposed AO algorithm based on PGM achieves the same achievable rate with a lower complexity compared to existing methods.

Index Terms—hybrid beamforming, millimeter-wave communication, intelligent reflecting surface (IRS).

I. INTRODUCTION

MILLIMETER wave (mmWave) communication is a very promising technology in 5G communication, it can achieve high throughput owing to relatively large bandwidth and beamforming technology. Unfortunately, the reflection and diffraction ability of mmWave is poor, which leads to its unstable performance in complex scenes [1]. However, the emergence of the intelligent reflecting surface (IRS) can alleviate this deficiency. The employment of the IRS can create more paths for communication, especially providing new available links when the mmWave direct link is blocked. So it can greatly improve its ability to maintain stable communication in a complex environment.

There have been many existing works on the IRS-aided mmWave system in recent years [2]–[5]. Reference [2] studied the weighted sum-rate maximization for mmWave multi-user system and jointly designed active beamforming (BF) at the base station (BS) and passive BF at the IRS. In Reference [3], the authors employed the hybrid beamforming (HBF) instead of full digital BF and used an alternate iteration (AO) algorithm to optimize the hybrid BF matrix and IRS phase shifts. Reference [4] further studied the joint design of the hybrid precoding/decoding matrix and IRS phase shifts by exploiting the sparse structure and large dimension of mmWave channels. However, [4] only considers single IRS case, and the low rank of the mmWave channel is not taken into account. Therefore, in Reference [5], the IRS composed of multiple sub-surfaces is used to avoid low rank of the mmWave channel to support multiplexing. However, the low rank still exists when the distance between the BS and the IRS is much larger than that among adjacent sub-surfaces. Reference [2] proved that the deployment of multiple IRS at

different locations could reduce the rank of mmWave channel and effectively enhanced mmWave coverage.

In this letter, we consider a multi-user multiple-input single-output (MU-MISO) mmWave wideband system aided by double IRSs and jointly optimize HBF and IRS phase shifts. Specifically, we derive a non-convex weighted sum-rate maximization problem, which is solved via the AO algorithm. We adopt the projection gradient method (PGM) to tackle the element-wise constant-modulus constraints for analog BF matrix and IRS phase shifts. Furthermore, the proposed algorithm is applied to the case that both analog BF and IRS use discrete phase shifts. Numerical results reveal the desirable performance for the proposed AO algorithm based on PGM.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System model

As shown in Fig.1, this letter considers an IRS-aided multi-user wideband system adopting orthogonal frequency division multiplexing (OFDM) to combat frequency selective fading, where a BS communicates with U single-antenna users. The BS is equipped with a uniform planar array (UPA) of N_t antennas and $N_{RF} \ll N_t$ RF chains. Note that HBF structure is a fully connected structure (FCS), where each RF chain is connected to all antennas. To be more realistic, we also consider partially connected structure (PCS) [6]. The mmWave channel is generally low-rank due to the sparse characteristics, so the spatial multiplexing capability of the system aided by single IRS is limited. To tackle this problem, we employ double IRSs in a distributed manner to create more paths. Each IRS is equipped with UPA, whose number of elements along the horizontal axis and vertical axis are equal, represented by N_{col} . So the IRS contains $N = N_{\text{col}} \times N_{\text{row}}$ elements, each of which can induce a phase shift to the incident signal independently by an IRS controller. To characterize the optimal performance of the IRS-aided mmWave system, we assume that the channel state information (CSI) in each channel coherence time is perfectly known at the BS based on the various wideband mmWave channel acquisition methods discussed [7].

The mmWave channel contains one LOS path and several NLOS paths with a corresponding relative time delay. Hence, the d -th delay tap of the delay domain channel of the BS-user u direct link is modeled as

$$\tilde{\mathbf{h}}_{d,u}^H[\mathbf{q}] = \sqrt{\frac{N_1 N_2}{L+1}} \alpha_d \sqrt{\frac{\mu D}{(1+\mu D)\kappa_1}} p(\mathbf{d}T) \mathbf{a}_{N_1} \mathbf{a}_{N_2}^H + \sum_{i=1}^{L-1} \frac{1}{(1+\mu D)\kappa_2} \alpha_i p(dT - \tau_i) \mathbf{a}_{N_1} \mathbf{a}_{N_2}^H \quad (1)$$

Magazine - Others

- ❑ Examples of magazines:
 - IEEE Communications Magazine
 - IEEE Software
 - IEEE Signal Processing Magazine
 - IEEE Security & Privacy Magazine
 - IEEE Pervasive Computing
 - IEEE Intelligent Systems
 - IEEE Internet Computing
 - Communications of the ACM,
<https://dl.acm.org/magazine/cacm>
 - ACM Queue, <https://dl.acm.org/magazine/queue>



Other Sources: Book Chapter

- ❑ A section or division of a book written by experts and focused on specific topic within a main research theme.
- ❑ Book chapters are compiled into a book → provides a balanced view and a perspective on research that is usually scattered in journals and other publications.
- ❑ Examples:
 - Artificial Intelligence and Machine Learning in Healthcare
 - Unmanned Aerial Vehicle Cellular Communications



2

A Survey of Air-to-Ground Propagation Channel Modeling for Unmanned Aerial Vehicles

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2.1 Introduction

Unmanned aerial vehicles (UAVs), also commonly referred to as “drones,” have long been used for military and specialized applications [25, 67, 115–117]. Owing to recent technological advancements, they have attracted major attention from industry for use (“use cases”) from package delivery to communications, surveillance, inspection, transportation, search and rescue, and others [1–6, 23, 42, 66, 96]. These UAVs can vary in size from small toys that fit in the palm of a human hand to large military aircraft such as the General Atomics MQ-9 Reaper (commonly termed Predator) [136], with a wingspan over 15 m. The small battery-powered toys can typically fly for up to 15 minutes, whereas the larger UAVs can be designed for long endurance (20 hours) and high-altitude operations (higher than 15 km). According to Goldman Sachs, there exists a \$100 billion market opportunity for UAVs in the commercial, civil government, and military sectors combined till 2020 [52].

Wireless connectivity with UAVs is a requirement for their integration into any national airspace and for facilitating new usages. However, UAV operating environments and scenarios introduce unique technical challenges, and these are presently being investigated by telecommunications companies such as AT&T [19], Vodafone [132], Ericsson [64], Nokia [18, 77], and Qualcomm [104], among others. A major challenge is obtaining realistic air-to-ground (AG) propagation models for various UAV operating environments and scenarios. Having accurate characterization of the AG channel is of paramount importance for designing robust and effective waveforms, modulation, resource allocation, link adaptation, and multiple antenna techniques.

The AG channel for UAVs has not been studied as extensively as terrestrial propagation channels. The available AG propagation channel models for high-altitude aeronautical communications generally cannot be employed directly for low-altitude UAV

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Other Sources: White Papers

- ❑ Provide information about a complex issue (product, service, or methodology)
- ❑ Often produced with marketing purposes.

Ericsson white paper
GPM-18/000530
November 2018



Advanced antenna systems for 5G networks

Recent technology developments have made advanced antenna systems (AAS) a viable option for large scale deployments in existing 4G and future 5G mobile networks. AAS enables state-of-the-art beamforming and MIMO techniques that are powerful tools for improving end-user experience, capacity and coverage. As a result, AAS significantly enhances network performance in both uplink and downlink. Finding the most suitable AAS variants to achieve performance gains and cost efficiency in a specific network deployment requires an understanding of the characteristics of both AAS and of multi-antenna features.



White Paper

Machine Learning in Certified Systems



DEEL Certification Workgroup
IRT Saint Exupéry
March 2021
Ref - S079L03T00-005



Understanding ZigBee RF4CE

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